

Aerodynamic design and evaluation of an open-nose supersonic drone

Eiman B. Saheby¹ , Anthony P. Hays² and Shen Xing, PhD³ 

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Abstract

The objective of this paper is to investigate the efficiency of a proposed supersonic drone configuration, in terms of drag, ram recovery, and fundamental flight performance factors. Sustainable supersonic cruise at Mach 1.6 is the major segment of the mission profile which affects the overall geometry resulting from the conceptual design phase, a tailless delta drone with an open-nose forebody, lofted around the inlet which consists of an analytical compression surface and a S-duct diffuser. Because the aerodynamics of this unconventional configuration is unknown, a series of CFD simulations using the ANSYS Fluent solver is coupled to the design process to predict both internal and external aerodynamics as a proof of the concept. The simulations indicate that the drone's overall drag is significantly lower than the other configurations with side or underside integrated inlets while the inlet pressure recovery is adjusted to maximize the engine thrust. The parasite drag at design speed is about 0.022 which is considerably lower than conventional configurations and the pressure recovery more than 0.96 is possible by applying the boundary layer bypass. A comparative study, with a developed thrust model, shows that the configuration satisfies mission requirements and exceeds them at transonic and supersonic flight phases.

Keywords

supersonic drone, delta configuration, open-nose concept, drag-polar, transonic-supersonic performance

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Introduction

The rise in the procurement of military UAVs by defense forces worldwide is one of the most significant factors projected to drive the growth of the UAV market.¹ Examples vary from a stealthy flying wing (such as the Northrop Grumman RQ-180) and high-endurance reconnaissance systems with large spans to micro drones. Most of the drone configurations in service are not designed as high-speed platforms. In fact there is no supersonic drone system in service with the ability to take-off and land on a runway. A series of papers (2006–2012) by Mizobata^{2,3} reports the design of small flight vehicles as a testbed for supersonic technology developments. From the same period, Kameda⁴ Murakami⁵ and Kazuhisa⁶ reported the JAXA's tailless delta drone with the top-mounted inlet as a low-boom (and low drag) drone technology demonstrator. In 2012, a concept of a light supersonic drone by Sibylle Walter⁷ presented another delta design with a pitot inlet at the nose to create a testbed for academic research such as thrust vectoring and control algorithms at Mach 1.40. A more recent concept of a heavy multirole supersonic drone is proposed by Eduardo Sepulveda⁸ (2019) as a 7 ton (empty weight) long range stealthy system with a serpentine inlet configuration

which required 93,000 N dry thrust (sea level) in order to supercruise at Mach 1.50 at 9.5 km altitude. The list of operational military unmanned flight systems consists of different types of cruise missiles, target drones, and subsonic reconnaissance and attack systems⁹ but no operational supersonic airbreathing drone with the ability for runway take-off and landing.

The foregoing discussion centers on a conceptual design and evaluation of a propulsion-based design supersonic drone proposed as an interceptor. The basic mission profile for such a system is shown in Figure 1. Analyzing the mission characteristics of the target is the first step in

¹ Assistant professor in the College of Aerospace Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing, China

² Aircraft Design and Consulting, College of Aerospace Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing, China

³ Ph.D. Supervisor in State Key Laboratory of Mechanics and Control of Mechanical Structures, Nanjing University of Aeronautics and Astronautics, Nanjing, China

Corresponding author:

Shen Xing, Supervisor in State Key Laboratory of Mechanics and Control of Mechanical Structures, Nanjing University of Aeronautics and Astronautics. 29 Yudao St., Nanjing 210016, China: shenx@nuaa.edu.cn

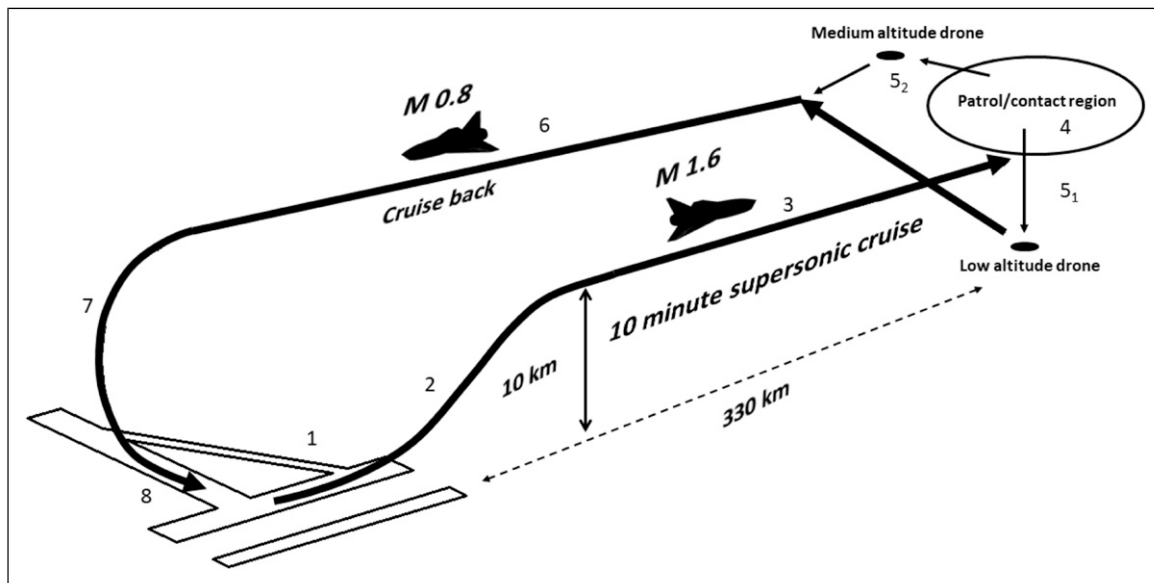


Figure 1. Mission profile for the supersonic interceptor drone.

Table 1. Mission profiles.

Mission segment	Altitude-time
1-Start-taxi-take-off	SL, 300 s
2-Climb	10 km, 420 s
3-Cruise to target area (AB-on)	10 km, 600–700 s
4-Maneuver, search, target designation	10 km, 600 s
5 ₁ -High-speed descent, system operation	10–1 km, 300 s
5 ₂ -Target change, system operation	1–10 km, 600 s
6-Gain altitude and cruise to base	10 km, 1,400 s
7-Descent + reserve time	10 km-SL, 300 s
8-Landing/taxi/shutdown	SL, 240 s

developing the design requirements for an interceptor. Speed, maneuverability, and flight altitude of the target UAVs determine the mission/system requirements. Heavy reconnaissance or strike UAV configurations fall into two categories: rectangular large span wings or stealthy blended (λ) wings with a low-profile dorsal intake placed near the leading edge. These systems are designed for subsonic and, in some cases, for high-subsonic (around Mach 0.8) flight regimes with low maneuverability so an interceptor system will have a significant advantage if it can fly supersonically ($M=1.6$ – 2.0) as a fast response system. The operational altitude of armed systems are generally below 10 km while some reconnaissance UAVs can fly at over 15 km⁹ so the minimum operational altitude for supersonic dash/intercepting is considered to be 10 km. Table 1 summarizes the mission segments, and Table 2 shows the minimum performance requirements for the baseline mission profile. Ground attack capabilities are not included in the mission requirements appropriate for the proposed interceptor drone although it is possible to adapt the planform for ground strike segments (like the F-14).

Table 2. Mission requirements.

Operational requirement	Value
Mission radius (minimum)	330 km
Supersonic dash (AB-on)	10 min
Cruise altitude	10 km
Subsonic cruise Mach No	0.80
Supersonic cruise Mach No	1.60
Estimated climb distance	9 km
Estimated descent distance	18 km
Supersonic dash distance	~300 km
Interception duration (AB-off)	15–20 minute
Supersonic turn (Mach 1.2 at 5 km)	5 g
Transonic turn (Mach 0.8 at 5 km)	5 g minimum
Take-off distance	0.5 km

Flying supersonically for a comparatively long period (10 min) highlights the requirement for propulsion system efficiency and aerodynamic performance; the classic propulsion technology on fighter jets relies on augmenting the thrust level by afterburner to overcome transonic drag rise which causes an increase in specific fuel consumption and limits the flight duration.¹⁰ Generally, if the overall supersonic drag of the flight vehicle remains below the maximum dry thrust level of the engine, then supersonic cruise without afterburner (or supercruise) is possible, and this corresponds to lower fuel consumption although it is also possible to accelerate through the transonic thrust pinch using the afterburner and then change to dry thrust for supercruise. When aerodynamic optimization (in the form of drag reduction) is limited by the other design requirements (such as stealth), engine thrust must be increased to cover supersonic drag penalties. The proposed drone configuration is designed based on an available engine on the

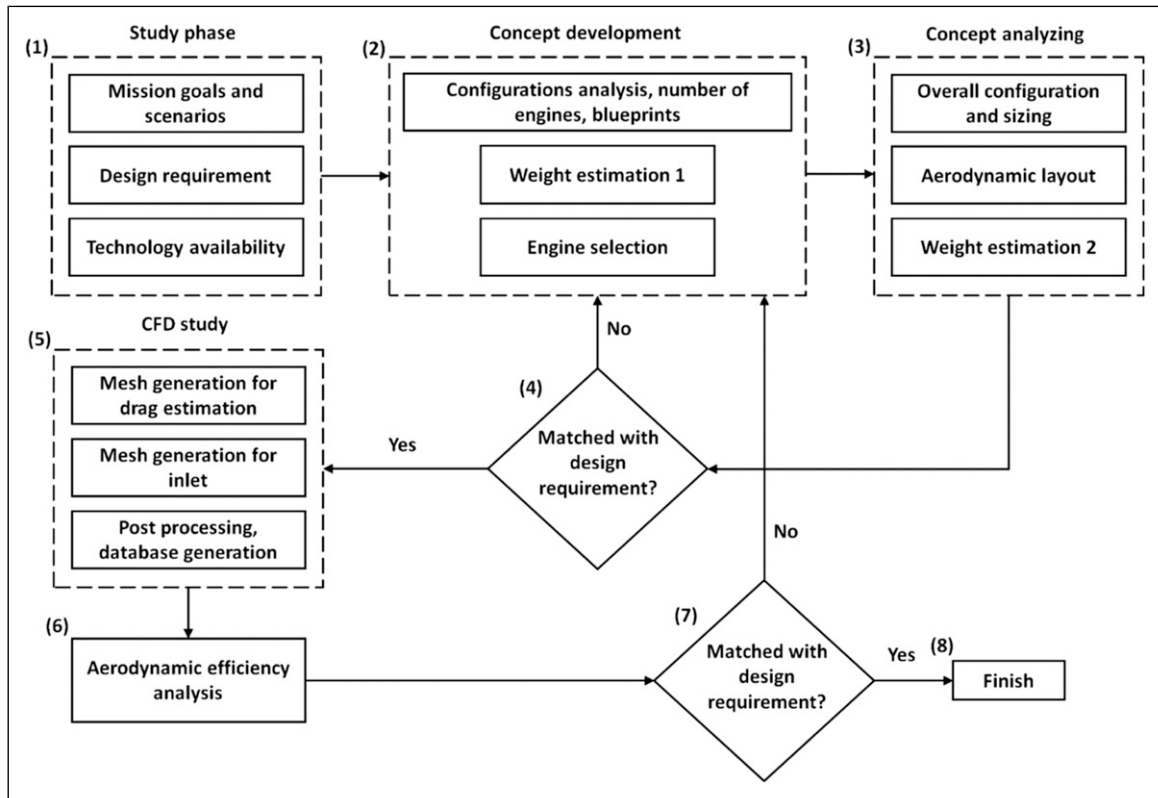


Figure 2. Conceptual design process.

market and its main performance factors are analyzed by a comparative study.

Initial sizing

Figure 2 shows the design flowchart following the first-order mass estimation method;^{11–14} estimating the maximum take-off weight (W_o), which is the sum of the payload (W_p), empty weight (W_e), and fuel weight (W_f). The initial empty weights W_e are estimated based on guessed values of W_o for the mission profile by the equation (1) and then the logarithmic trend (equation (2))¹⁵ is used to determine an identical answer

$$W_e = W_o - W_f - W_p \quad (1)$$

$$\log W_o - B \log W_e = A \quad (2)$$

The regression coefficients A and B are determined based on applying the least square method to find regression coefficient of a logarithmic relation between W_e and W_o of a similar fighter configurations. The relationship between fuel fraction and W_o is determined by consumption during the mission profile, using semi-empirical analysis. For this step the L/D at cruise was estimated to be about 10 with the internal payload and 3 at supersonic phase. The maximum payload mass for the system is 1,100 kg, but for the supersonic mission profile, four hardpoints under the wing carry short range missiles with a maximum weight of 100 kg (with a tolerance of about 20 kg). Figure 3a shows results for three different

sets of regression coefficients. Although sizing the concept by this method is not accurate especially when no comparable drone exists, according to mission profile and estimated weights, the configuration should be similar to a light fighter (or trainer). As it will be shown in section 2.3, the weight estimations based on components weight (process 3 in Figure 2) shows that third design, with regression coefficient of 0.83 and 0.87 in Figure 3a, is a better estimation. Light fighters (or trainers) are not designed for sustained supersonic flight so that without similar designs, it is hard to estimate the aerodynamic parameters accurately, but the engine weight and dimensions can be determined because the selections are limited by the market; engines with 30–50 kN dry thrust have been adapted to power heavy combat UAVs such as the BAE Taranis or X-47 prototypes.¹⁶

Table 3 summarizes the characteristics of some well-known systems capable of adapting on heavy drones. By considering weight, length, and diameter, the Snecma M-88 appears to be a better choice although the General electric F-404 and Rolls Royce Adour family have been used on many flight vehicles. By considering an engine like the M-88, the mass budget for the engine (with afterburner) is about 1,100 kg with maximum dry thrust of 50 kN at the sea level. Figure 3b shows the drone thrust to weight ratio compared with other well-known interceptors.

The design space of a complex system is determined from consideration of many factors but the required stealth level, aerodynamic performance, and propulsion system integration affects the shape of flight vehicle more than others. These fundamental factors and their relation to

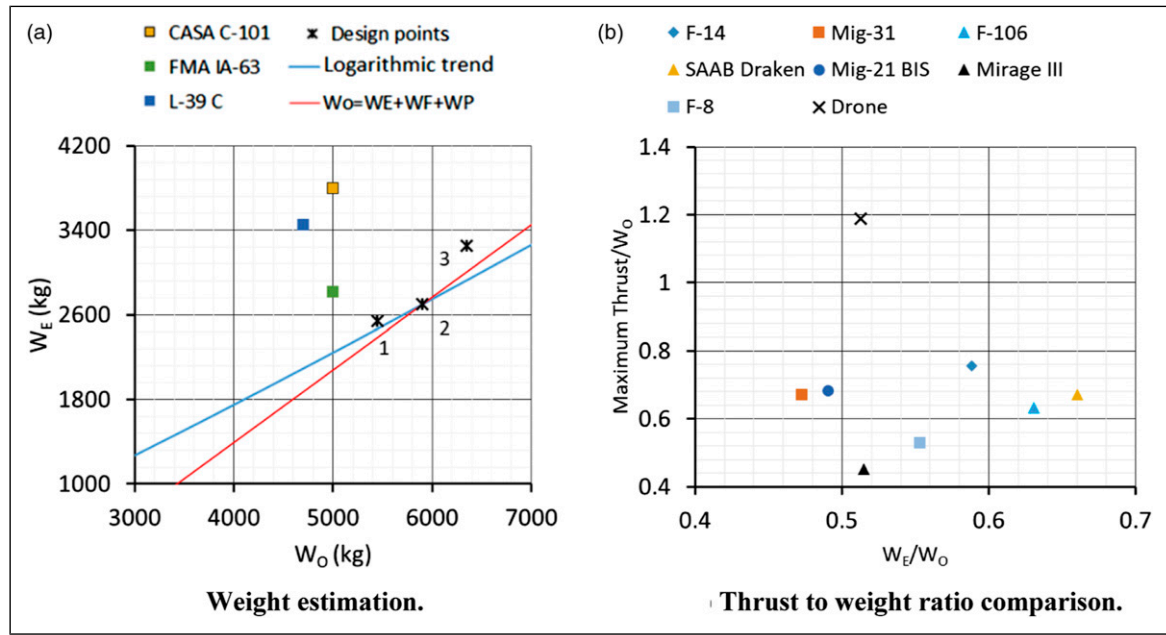


Figure 3. Result of initial sizing. (a) Weight estimation. (b) Thrust to weight ratio comparison.

Table 3. Engine characteristics.

Engine	Dry weight (Kg)	Dry thrust (kN)	Length (mm)	Diameter (mm)
R-R Turbomeca Adour	809	27	2,900	570
Honeywell FI24	521	28	2,590	914
Klimov RD-33	1,055	50	4,229	1,000
General Electric F-404	1,035	48.9	3,900	890
Snecma M-88	897	50	3,540	696
R-R RB199-104	979	40	3,600	720

Table 4. Range comparison.

Airplane	Internal fuel capacity (kg)	Engine	Thrust to weight ratio	Combat range (km)
F-35 A	8,270	P&W FI35	1.08	1239
F-16 C	3,200	P&W F-100-PW-229	1.09	1370 (1,000 kg payload)
Saab JAS 39	3,000	Volvo RM12	0.94	800
Mirage 2000	3,100	Snecma M53-P2	0.88	1475 (1,000 kg payload)

mission scenarios are reviewed here for selected fighter configurations (but explaining the details for each study case is out of the scope of the current paper). A tailless delta was selected for the drone's baseline design based on the following analysis.

Stealth level

Stealth is a crucial defensive ability for strike aircraft against interceptors or detection by ground radar. When the strike aircraft cannot react to the interceptor, as is the

case for a strike UAV, then this advantage may not be a priority in interceptor design. In our case, UAVs without air-to-air combat ability are not a threat to interceptors and being stealthy may only provide a better surprise element for the interceptor. The required subsystems and materials for a stealth system increase software/hardware complexities, weight, and costs.¹⁷ The stealthiest designs (such as the F-22) use a combination of radar-absorbent coatings, surfaces, and edges that reflect radar signals at discrete angles, and hidden compressor face and nozzles. Each of these elements results in a level of stealth in

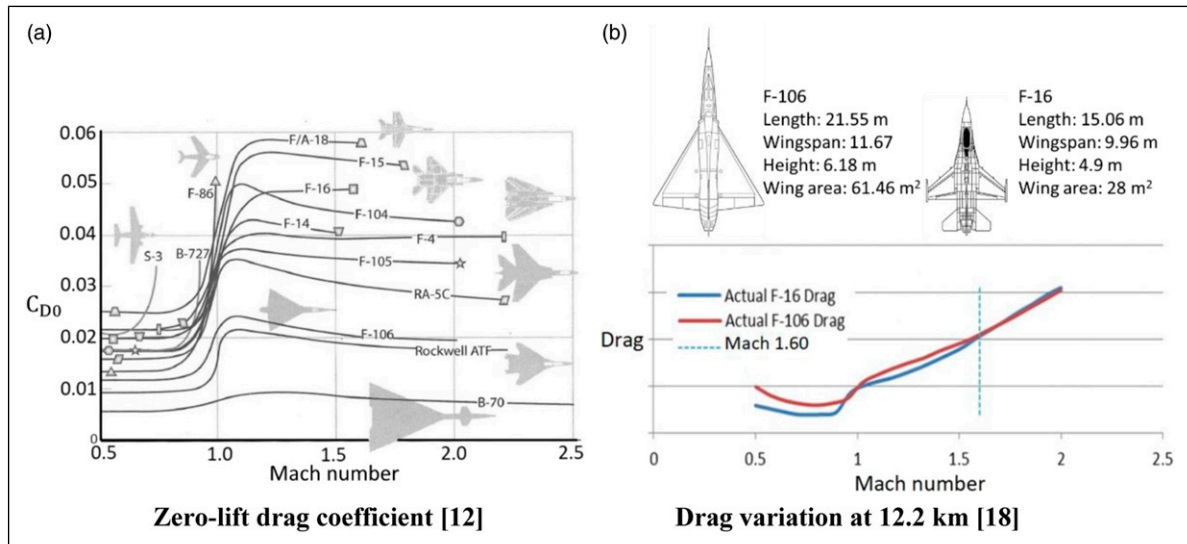


Figure 4. Aerodynamic characteristics and inlet shape. (a) Zero-lift drag coefficient.¹² (b) Drag variation at 12.2 km.¹⁸

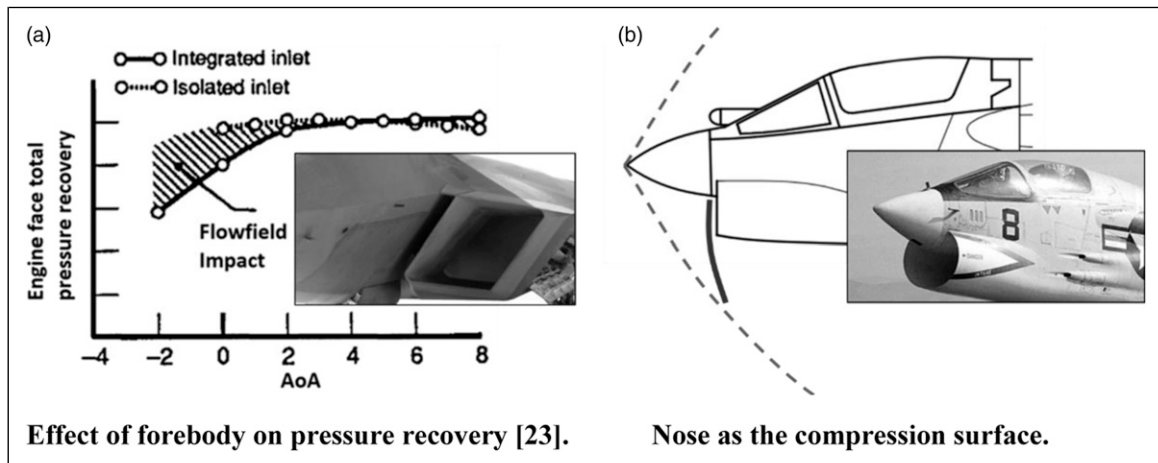


Figure 5. Side integrated and open-nose inlets. (a) Effect of forebody on pressure recovery.²³ (b) Nose as the compression surface.

design, but when a highly stealthy configuration is not required, it is still possible to include some stealth elements which do not degrade performance, such as hiding the compressor face using a curved diffuser duct as in the current design.

Aerodynamic/propulsion integration

The wide usage of a tailless delta (or delta + canard) configurations with side-mounted inlets in European fighter projects prove its effectiveness for adapting jet engines with lower thrust level (corresponding to lower overall drag) compared with other configurations. A comparison of thrust and range is summarized in Table 4 where the Mirage 2000C, which is primarily an interceptor, shows its aerodynamic efficiency with a better range and lower thrust engine compared with the F-16. The F-35 has a considerably higher T/W with A/B than the Mirage 2000 although the Mirage 2000 has a higher maximum Mach number. This suggests that stealth imposes a significant drag penalty.

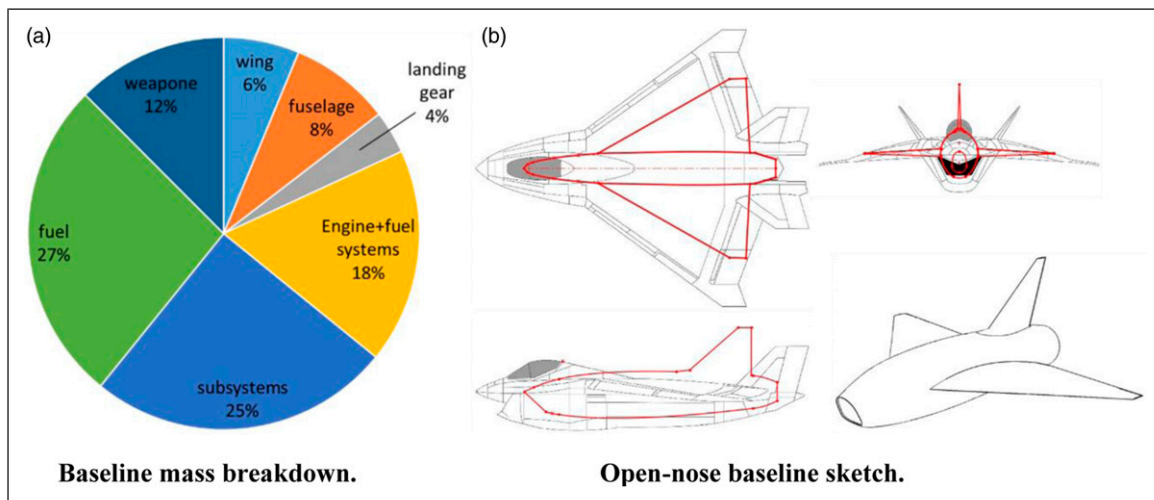
Brandt's article¹⁸ (2018) discussed the possibility of drag reduction for a stealth configuration by allowing both engine and airframe sizes to vary as the two are optimized together, but it is difficult to reduce the drag below that of area-ruled tailless delta configurations such as the F-106 or Mirage 2000. As Figure 4a shows, the combination of slender fuselages and delta wing results in a much lower parasite drag coefficients. That is why the supersonic drag of a larger airplane like F-106 (with 21.5 m length and wing area of 61.5 m²) is almost equal to F-16 (with 15 m length and wing area of 28 m²) which is demonstrated in Figure 4b.

For inlet propulsive efficiency of a single engine airplane, the side-integrated concept is not as efficient as underside or chin inlet configurations.¹⁹ Side-mounted inlets add higher free volume for subsystems but the Y-shape duct (twin duct) produces higher turbulence levels, asymmetric flow problems, lower pressure recovery at sideslip, and boundary layer diverter drag.^{20,21}

For a fighter jet with the capability of short supersonic flight, reduction of inlet pressure recovery by shockwaves

Table 5. Interceptor drone design considerations.

Criteria	Requirement/Considerations
Overall configuration	Single engine tailless delta with the open-nose configuration
Landing gear	Tricycle
Radar/sensor integration	Active scan is necessary. The forebody should provide enough space for a flat panel array radar and its subsystems
Hardpoints	4 external hardpoints. Tips and midwings
Supersonic response payload	Short range passive air-to-air missile with maximum weight of 100 kg (and no gun)
Maximum payload	1,300 kg
Engine	40–50 (kN) fighter class low bypass turbofan. (R-R Turbomeca variants or Snecma M-88)
Stealth level	Medium; radar wave absorbing material and hiding the compressor face by duct curvature

**Figure 6.** Result of conceptual design. (a) Baseline mass breakdown. (b) Open-nose baseline sketch.

and viscous interactions may be tradable with other performance factors, but for a flight system designed for sustainable supersonic cruise as for this design, any small reductions in pressure recovery corresponds to considerable thrust loss.²² In addition to the shockwaves on inlet compression surfaces, the forebody section affects the captured streamtube especially at supersonic regimes. Figure 5a demonstrates the forebody adverse effects due to the integration of a side-mounted caret inlet at $M_\infty=1.8$.²³

Generally, when $1.5 < M_\infty < 2.5$, the pressure recovery losses by the forebody are considerable.²⁴ Isolating the inlet entrance by designing the forebody based on the inlet is an established solution to these problems. Open-nose forebodies based on pitot or axisymmetric spike inlets (like the F-100 Super Sabre and Mikoyan MiG-21) are examples of this solution but a better design can be seen on the chine configuration of F-8 in Figure 5b. The poor volume efficiency at the nose section made axisymmetric cone forebody less suitable for integrating a large radar, but for the chine configuration, this problem is lessened. Removing the diverter drag by using the nose as the compression surface and efficient flow diffusion with less distortion at the compressor face also make the open-nose concept an attractive selection for the supersonic drone design.

Table 6. Characteristics of the baseline concept.

Total weight (based on maximum payload 1,100 kg)	6,330 kg
Empty weight	3,200 kg
Required fuel	1,835 kg
Drone length	10.6 m
Wing span	6.70 m
Height (bottom to the tip of the rudder)	1.49 m
Duct length (from the nose to the compressor)	5.80 m
Wing area	27.90 m ²
Length of the supersonic compression surface	0.94 m
Inlet diameter (at the compressor face)	0.75 m
Base diameter	1 m
Wing root	6.18 m

Baseline configuration

The result of the initial study and design tasks in previous sections was to develop a realistic design space for the drone concept based upon the available propulsion system characteristics for Mach 1.60 + cruise. Primary design requirements are summarized in Table 5, resulting in the

Table 7. Components and mass breakdown.

	Component	Property/considerations
1	Removable composite nose	30 kg composite material
2	Radar	100 kg
3	Duct	100 kg based on fiber carbon composite
4	Electronics	150 kg
5	Main fuel tank	0.90 m ³ (Plastic 22 kg empty)
6	Secondary fuel tanks	3 × 0.22 m ³ (Plastic 5 kg empty)
7	Wing fuel tank	2 × 0.12 m ³
8	Engine (and subsystems)	1,100 kg (Engine+reheat pipe+nozzle+pump)
9	Wing composite section	100 kg
10	Wing aluminum section	400 kg (Connector+aluminum section of the wing)
11	Main landing gear	200 kg undercarriage
12	Nose landing gear	40 kg
13	Hydraulic oil tanks	15 kg
14	Fuselage (skin and internal structure)	600 kg based on aluminum 2024 and 25% composite
15	Vertical tail	90 kg based on aluminum 2024
16	Hydraulic subsystems	45 kg
—	Empty weight	2,997 kg (3,200 kg estimated)
—	Total weight	5,515 kg (6,300 kg estimated)

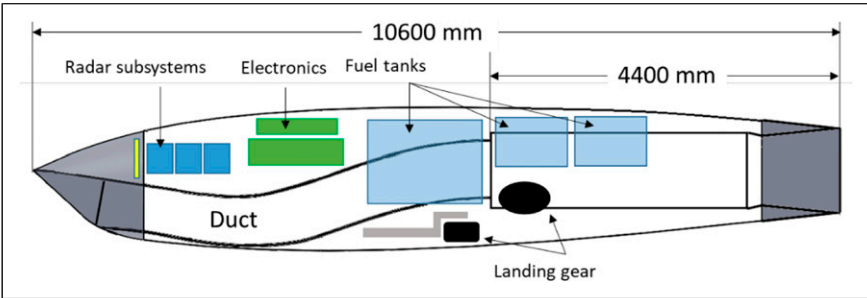


Figure 7. Schematic of fuselage lofted around the propulsion system.

selection of a combination of open-nose fuselage with the pure delta wing.

For the aerodynamic design and engine integration requirements, the fuselage of the drone is designed by distributing the fuselage surface around the induction system (captured streamtube) similar to MiG-21 which results in a low drag configuration, but instead of a spike inlet, an open-nose (or chine) inlet is adopted with cross-sections varying from elliptical (at the nose) to cylindrical. Figure 6b compares a three view of the drone with that of the X-32, and Figure 6a illustrates the mass distribution. The required volume of the cockpit, radar, and avionics subsystems results in a large and odd forebody for the X-32, but for the drone concept in Figure 6, the nose section can be designed by extending the outer (external) surface of the supersonic diffuser because there is no need for a cockpit and its subsystems.

Determining the aerodynamics of the concept is crucial; not only the overall drag but also the inlet efficiency is unknown; therefore, this analysis integrates CFD simulations to assess the aerodynamic efficiency and to compare the overall performance factors. Table 6 summarizes the

main dimensions after second weight estimation. The weight estimation II in concept analysis flowchart (process 3 in Figure 2) is summarized in Table 7, and it determines a more realistic mass estimation/distribution by adding the weight of the main elements to the CAD layout. The methods of initial length sizing are not standardized, but for the current concept, dimensions are estimated by trends from known light fighter or trainer systems.

Aerodynamic layout design

In this section, the main aerodynamic surfaces are designed based on the constraints from other subsystems to create a realistic configuration for the CFD simulations.

Forebody/inlet/nozzle

In the propulsion-based-design approach, the structure of the propulsion system (inlet, engine, and nozzle) determines the fuselage architecture and affects the subsystems integration.²⁵ As Figure 7 shows, the fuselage is

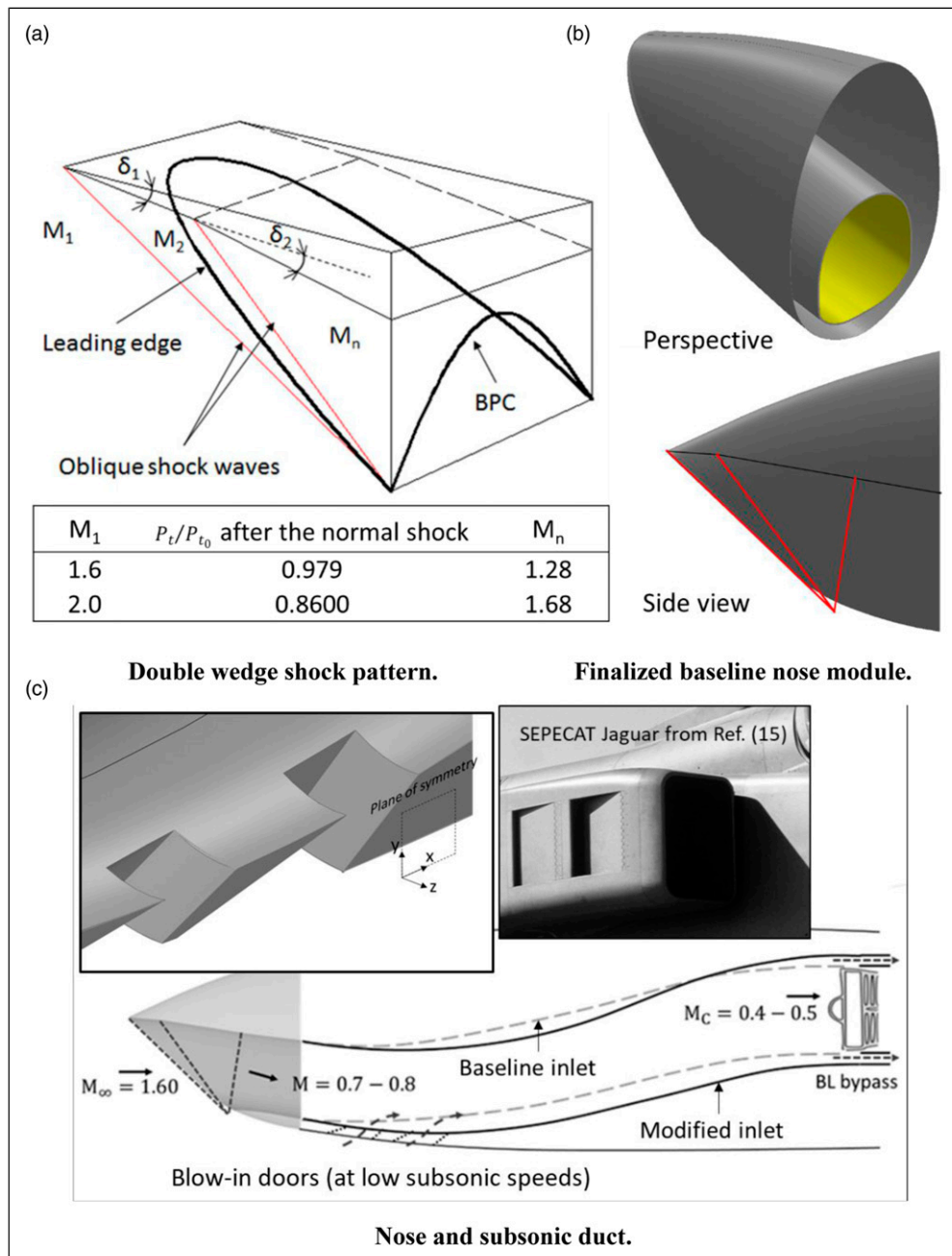


Figure 8. Open-nose forebody. (a) Double wedge shock pattern. (b) Finalized baseline nose module. (c) Nose and subsonic duct.

generated by the geometry transition between the elliptical cross-sections of the nose to the circular cross-sections (gray color) at the base plane (nozzle). Nose design is determined by the compression surface and throat area. These elements are designed by selecting the number of oblique shockwaves (3-shocks system is selected), terminal shock (throat) Mach number (M_n is the Mach number before the normal shock), and maximum compressor mass flow rate. Figure 8a demonstrates how the compression surface is derived from the shock pattern of a double wedge shock generator. In the shock-based design method, the base-plane curve (BPC in Figure 8a) prescribes the compression surface while the external surface should be designed based on the overall configuration (forebody requirements) of the flight vehicle.^{26,28} For

wedge-derived surfaces, the generating flowfield is defined by a two-dimensional oblique shockwave equation.^{27,28}

Applying the isentropic compression surfaces, derived from conical shock patterns are reported for low-supersonic flight systems too,^{29,30} but these curved surfaces are more useful at high-supersonic or hypersonic speeds.³¹ Figure 8b shows the nose segment. The composite structure has two roles: intake and radar radome. The lip profile with a sharp curve on the compression side and a rather blunt profile on the cowl side couples the internal surface of the inlet to the external surface of the forebody. The sweptback cowl geometry follows the first oblique shock front to generate less drag.

The subsonic duct is determined by the throat cross-section of the supersonic diffuser, a compressor face

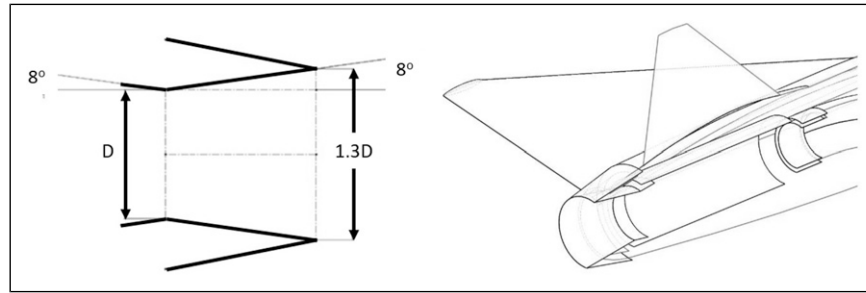


Figure 9. Nozzle layout at expanded condition.

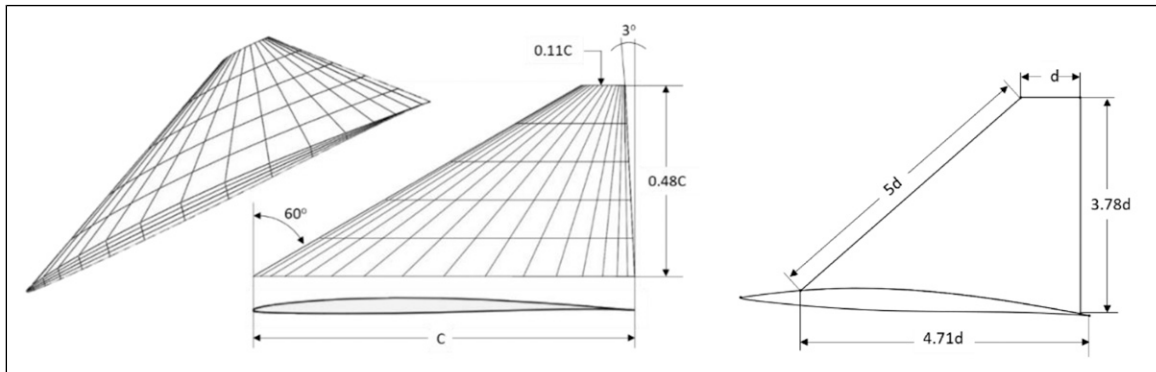


Figure 10. Wing (left) and vertical stabilizer (right).

offset, and a centerline to control the geometry transition between the throat and the compressor face. The control points of the centerline spline are coordinated based on required volume for the radar, electronics, and the nose landing gear. The inlet must support the compressor at the start and take-off phases. The fixed geometry of the diffuser is designed for supersonic performance so that, like many inlets with similar operating requirements, auxiliary doors may be required to increase the captured mass flow at the take-off condition. These doors are added to the forebody after initial simulations and measurement of the mass flow rate at the outlet at static operation condition. Figure 8c shows the location of these doors copied from the arrangement on the SEPECAT Jaguar inlet. Accurate design of auxiliary doors is an optimization problem requiring many design-simulation iterations and it is not considered in this project. The total (projected) internal area of the auxiliary doors in Figure 8c are 34% of AIP (outlet) area.

Another challenge related to the diffuser is boundary layer management. It is possible to suck out the boundary layer at the end of the subsonic diffuser by installing a porous zone or placing a gap between the compressor and duct. The impact of boundary layer bleed was investigated by redesigning the duct and adding a concentric gap. Figure 8c shows two different diffuser ducts used in these studies. The concentric bleed ring at the outlet is designed based on the maximum thickness of the boundary layer at the end of the duct.

Using the boundary layer bleed at the shock root is challenging because the shock root (potential location of a porous suction zone) is located on the upper surface of

Table 8. Wing parameters.

Airfoil name	NACA 63204 a=1.0 (root and tip)
Thickness/Chord ratio	0.04
Ideal CL	0.24
Twist Loc	0.25

the diffuser and below the radar antenna; although with a transonic Mach before the terminal shock and a thin boundary layer on the compression surface, the boundary layer-normal shock-interactions are not strong enough to generate a considerable flow separation.

The last segment of the propulsion system is the variable axisymmetric nozzle as shown in Figure 9. When afterburning engine operation is required, complex variable-geometry nozzles must be used. In the current concept, a symmetric nozzle structure at expanded (full afterburner condition) is used to estimate the upper limit of drag. The nozzle parameters and design methods are well addressed in textbooks, such as Ref. 14.

Wing/vertical stabilizer

The wing is designed by considering the aspect ratio of some tailless fighter jets. Figure 10 shows the resulting geometry using Open VSP software based on 6-series airfoil (similar to the F-16 wing airfoil) and 1.5° washout. The importance of a delta wing configuration is related to the wing's transonic and supersonic efficiency,^{32,33} natural simplicity in manufacturing, low wing-loading, and sturdy

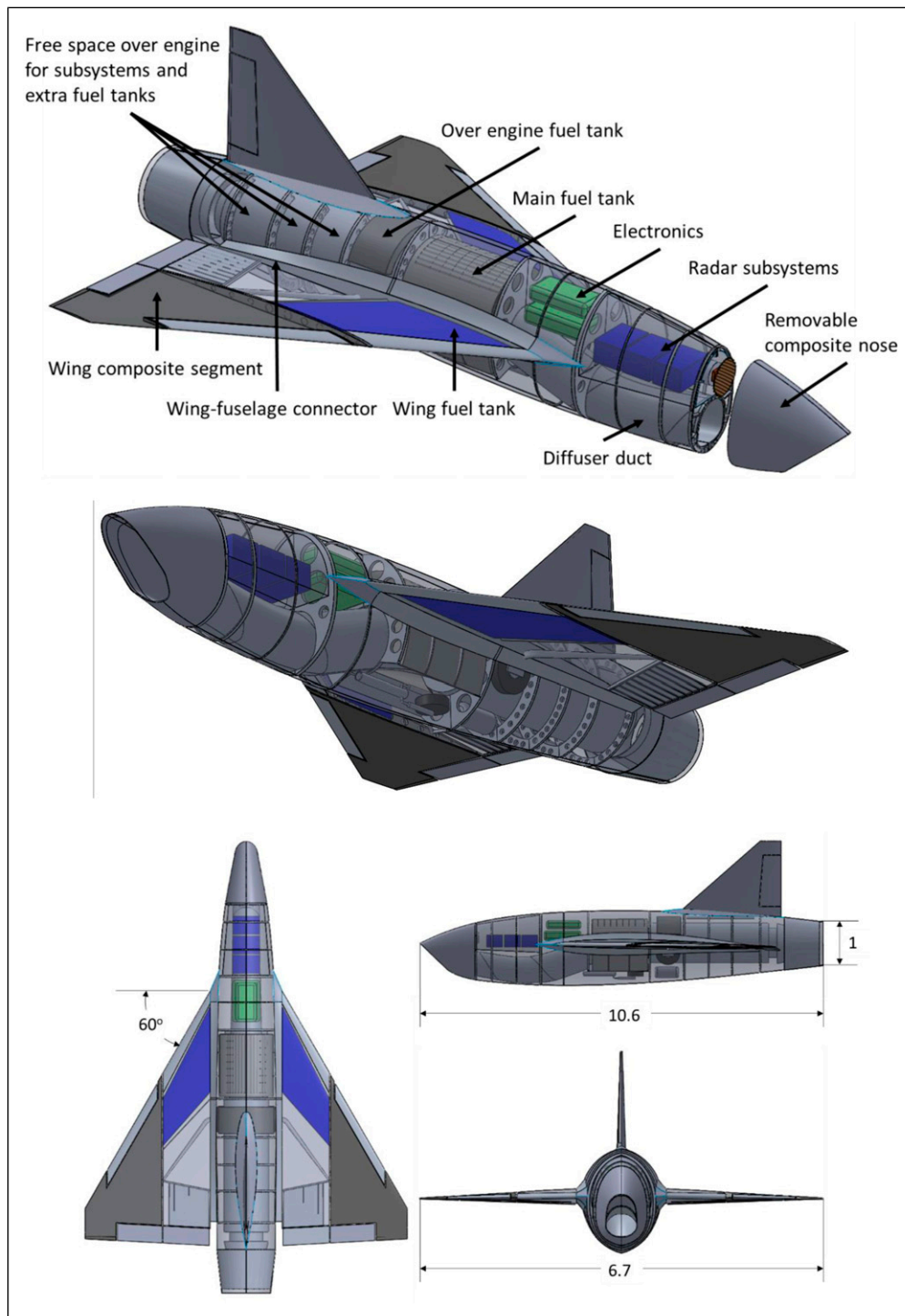


Figure 11. Drone CAD layout (baseline).

structure which outweigh its disadvantages such as lower lift coefficient at low subsonic speeds and handling problems. For the current design, the higher maneuverability of tailed configurations is not required, so a pure tailless delta can be an appropriate selection. Table 8 summarizes the wing parameters. The finalized wing layout merges with the fuselage by a wing fillet. This surface can be seen in Figure 11 which demonstrates the aero-layout and the main subsystems. Design and analysis of the control surfaces are not covered in this paper. The

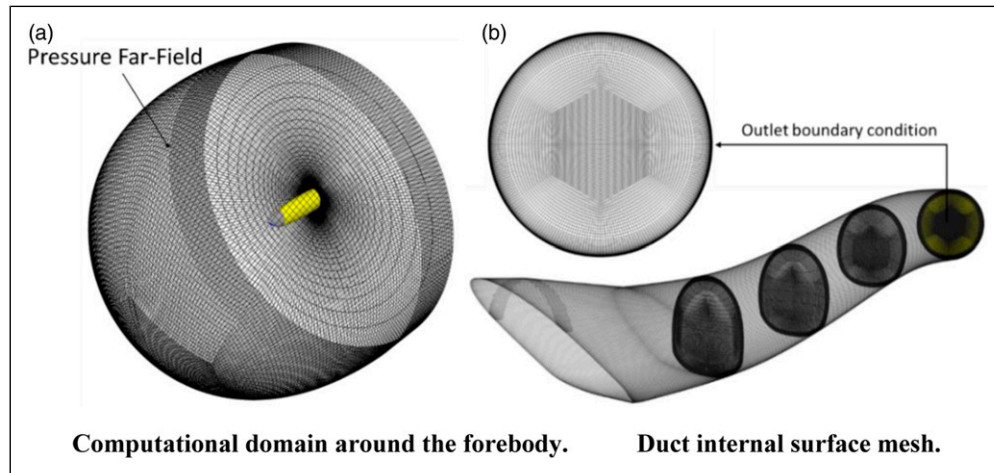
wing and vertical stabilizer are modeled without active control surfaces for CFD simulation. The vertical stabilizer has a symmetrical airfoil and its dimensions are selected from a trend of area ratios of available configurations.

The finalized layout for simulations

Once the propulsion system and its external dimensions are determined, the fuselage can be sized by considering the main subsystems volume. Figure 11 shows the final

Table 9. Atmosphere condition.

Altitude (km)	Speed	P_{∞} (pa)	T_{∞} (K)
Sea level	Low subsonic and high AoAs	100,000	298.15
5	Maneuver at high subsonic to transonic	54,048	255.67
10	Supersonic dash boundary (Reynolds number 1.3595×10^7 at Mach 1.60)	26,500	223.25
18	Mach 2^+ studies	7565.2	216.65

**Figure 12.** Mesh structure of inlet-forebody for design and off-design conditions. (a) Computational domain around the forebody. (b) Duct internal surface mesh.

CAD layout of the drone in more detail. According to the engine dimensions and the required space for fuel tanks, applying the area-rule on this configuration is challenging. Any reduction in fuselage volume around the engine limits the fuel tank capacity. From Ref. 14, it can be found that the fuselage fineness ratio varies from 3 (for subsonic aircraft) to 14 (for supersonic aircraft) for drag minimization. The fineness ratio (the ratio of the length of a body to the diameter of a circle whose area is equal to the maximum cross-sectional area of the body) of our concept is around 6.1, but this classic ratio cannot be compared with other fuselages with integrated inlets because the fuselage design is strongly dependent on the inlet. The rear section of the fuselage is configured to permit an easy modification to accommodate an engine with compressor face diameter up to 0.95 m. This tolerance was necessary for initial aerodynamic layout because the drag and required thrust level was unknown for this configuration.

In contrast to the fuselage integrated inlets which are modifiable, changing the engine type needs redesigning the supersonic diffuser which corresponds to change in the forebody and its drag. Now, if the new engine cannot overcome the resulting drag, then a design loop problem would be formed. To solve this problem, some safety margins were considered for the engine casing and compressor face diameter; if the compressor face diameter is reduced (by changing the engine) after the thrust-drag study, then the forebody drag will not increase.

Computational methodology

The internal and external aerodynamics of the system is simulated by solving the Navier–Stokes equation (N-S) coupled with the k- ω turbulence model in fully structural cell domains to predict velocity and pressure gradients. The vector form of the three-dimensional N-S equations, coupled with the k- ω turbulence model³⁴ may be written as

$$\frac{\partial E}{\partial x} + \frac{\partial F}{\partial y} + \frac{\partial G}{\partial z} + \frac{\partial q}{\partial t} = \frac{\partial E_v}{\partial x} + \frac{\partial F_v}{\partial y} + \frac{\partial G_v}{\partial z} + S \quad (1)$$

where E , F , and G are inviscid flux vectors, and E_v , F_v , and G_v are viscous flux terms. q is a variable vector, and the S is a source term vector related to the turbulence model. The N-S and applied turbulence model in this project are well-known in the computational gas dynamics; therefore, their extensive formulations and related explanations are not presented in the current draft. The static pressure and temperature for current calculations are derived from the standard atmosphere and summarized in Table 9.

Forebody/inlet mesh

Because the inlet flowfield is not affected by the wings and fuselage, an isolated inlet-forebody model as shown in Figures 12a,b can be used for discretizing the governing equations. Using this approach, the computational cost is reduced and the mesh density is concentrated on the internal volume. Figures 13a,b show the computational domains for static operation on the ground when the auxiliary doors are open. The major challenge in applying the

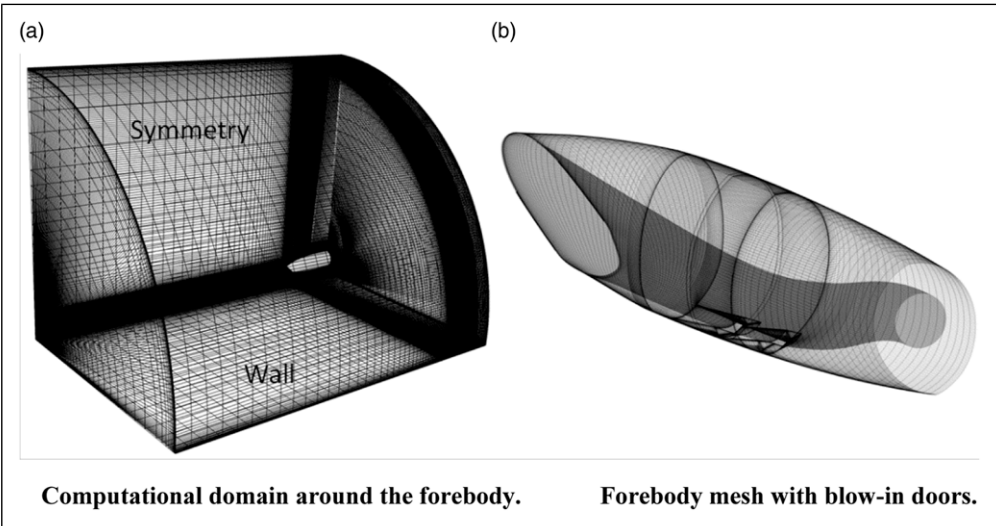


Figure 13. Mesh structure of inlet-forebody for static operation. (a) Computational domain around the forebody. (b) Forebody mesh with blow-in doors.

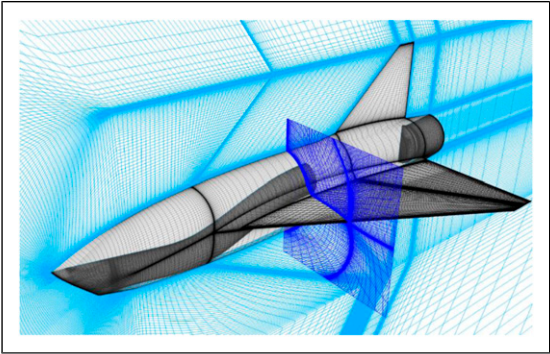


Figure 14. Half drone mesh structure.

structured cell with a high orthogonality factor is related to the bluntness of the cowl lip and leading edge of the compression surface. This problem is solved by expanding the cell domains to a spherical boundary, which is shown in Figure 12a.

Drone mesh

For the drone configuration, the mesh density is focused on the main external aerodynamic surfaces and a lighter grid is applied on the inlet duct surface. As Figure 14 shows, half of the drone is used for mesh generation based on the plane of symmetry boundary condition. For all models, the geometry is surrounded by a pressure farfield boundary condition. The pressure farfield is connected to a pressure outlet at the end of the diffuser, and for the complete model, a mass flow inlet is applied to the end of the nozzle which can be seen in Figure 14. The computational domain of the drone is large enough to capture the pressure gradient around the forebody at low subsonic speeds where the continuum effects are dominant. In order to start the solver, the values of static pressure, temperature, and Mach number from the pressure farfield are

Table 10. Result of grid study for the S-duct inlet.

	Grid	Cell number ($\times 10^6$)	Index	
Inlet	Light 1	1.2	P_t (AIP) at $M_\infty=1.6$ (Pa)	316,419.59
—	Light 2	2.3	—	318,990.10
—	Fine	3.9	—	319,161.72
—	Ultra-dense	29.1	—	320,114.07
Inlet+aux	Light	3.3	—	194–203
—	Fine	5.6	k_{ave} (m^2/s^2)	201–210
—	Dense	17.7	—	215–219
Drone	Initial	6.4	C_D at $M_\infty=1.6$	0.0228
—	Dense	16.4	—	0.0234
—	Ultra-dense	26.6	—	0.0239

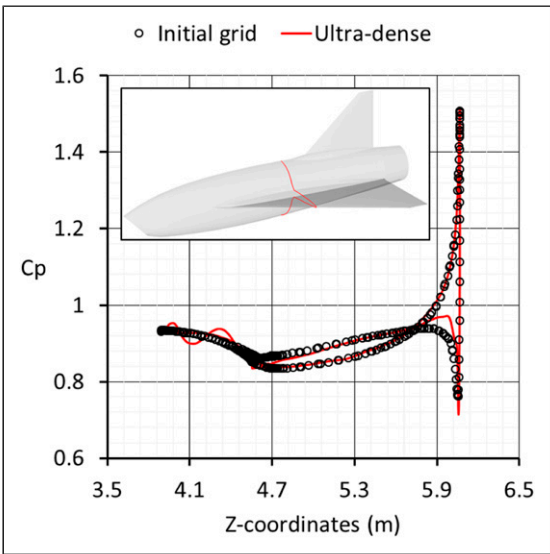


Figure 15. Pressure coefficient distribution on a cross-section.

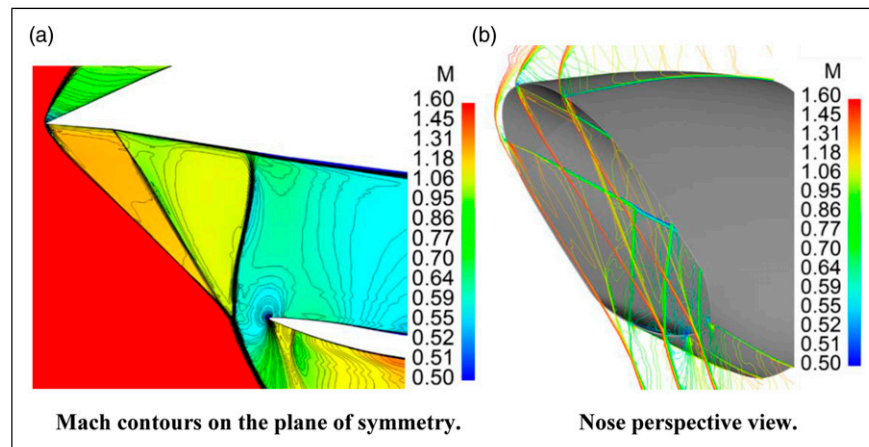


Figure 16. Shock waves structure at subcritical operation, $M_\infty=1.60$, $\alpha=0.0^\circ$, $\beta=0.0^\circ$. (a) Mach contours on the plane of symmetry. (b) Nose perspective view.

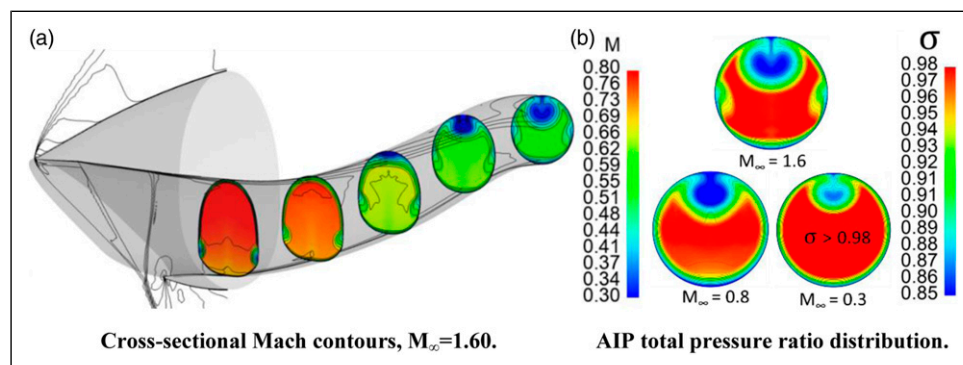


Figure 17. Inlet flow structure, $\alpha=0.0^\circ$, $\beta=0.0^\circ$. (a) Cross-sectional Mach contours, $M_\infty=1.60$. (b) AIP total pressure ratio distribution.

used for domain initialization. The cell growth rate for the internal surface of the inlet is 1.10 although the mesh around the nose is refined by regional adaptive module of the ANSYS Fluent.

Grid study

The multiblock domains are solved by a desktop with 40 computational nodes using the ANSYS Fluent solver. The solution starts with the first-order discretization algorithm, and the iterations stops when the residuals exceed 10^{-5} , then the discretization method is changed to the second order and iterations continue until the solution convergence, which is the order of 10^{-4} for the second-order scheme. Table 10 summarizes the grid convergence study. The required cell number for the inlet is determined by comparing the averaged total pressure ratio P_t at the AIP, and for the drone, the drag coefficient is the index. Figure 15 demonstrates the pressure coefficient distribution on a (transversal) cross-section for the initial grid and ultra-dense case. Although the ultra-dense mesh captures a detailed pressure distribution (especially on the fuselage), the resulting drag coefficients varies about 4.8% comparing to the initial grid.

Results and Discussions

Inlet aerodynamics

A detailed investigation of the inlet's flow structure and its efficiency was documented in a separate publication with more details³⁴ in 2021. But briefly, at the supersonic design point, the nose compression surface of the forebody generates shock waves which reduce the supersonic flow from Mach 1.6 freestream to 1.28 before the terminal shock wave as was predicted analytically in the section Forebody/Inlet/Nozzle. Figure 16a illustrates the shock waves by the Mach contours on the plane of symmetry at subcritical operation modes, and Figure 16b demonstrates the three-dimensional shock pattern by Mach contours on the parallel longitudinal cross-sectional planes.

Without any bleed or flow control system, the flow in the S-duct is significantly affected by boundary layer separation along the duct centerline and the development of the secondary flows which are well investigated phenomena in inlet gas dynamics.^{35,36} The cross-sectional Mach contours in Figure 17a and total pressure ratio (σ) distribution on the AIP in Figure 17b demonstrate the phenomenon. The total pressure recovery ratio of a few high-performance inlets, mainly designed for Mach 1.20–1.80^{37–39} are compared in Figure 18. The gulfstream

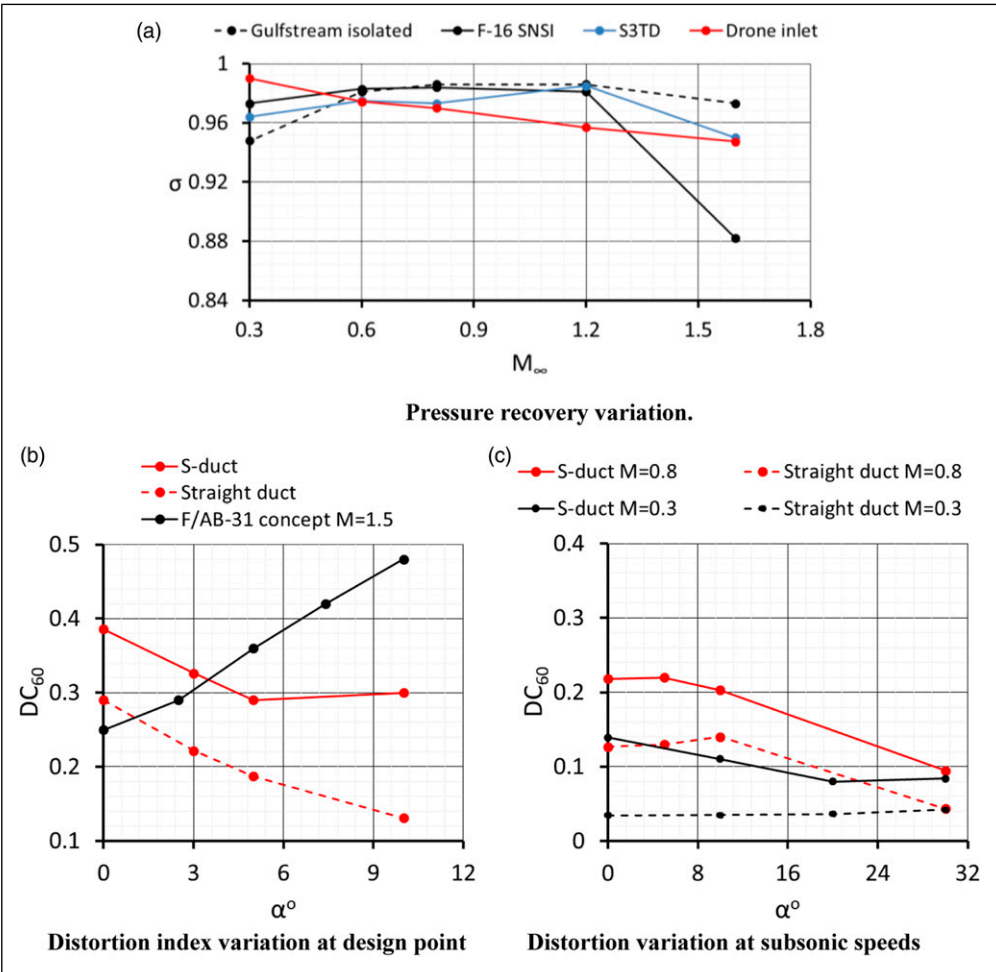


Figure 18. AIP's flow parameters without BL gap. (a) Pressure recovery variation. (b) Distortion index variation at design point. (c) Distortion variation at subsonic speeds.

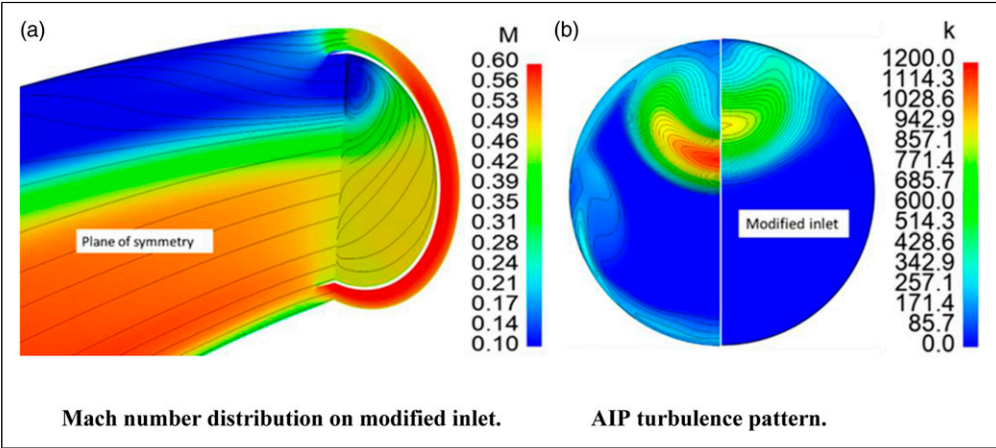


Figure 19. AIP pressure recovery at design point. (a) Mach number distribution on modified inlet. (b) AIP turbulence pattern.

Table II. Efficiency factors of modified inlet.

Compressor face Mach number	Altitude	M_∞	σ	k_{avr}	M_{bypass}
0.411	10 km	1.6	0.967	134.97	0.59
0.453	10 km	1.6	0.953	231.86	0.58
0.454	8 km	1.38	0.979	127.31	0.52

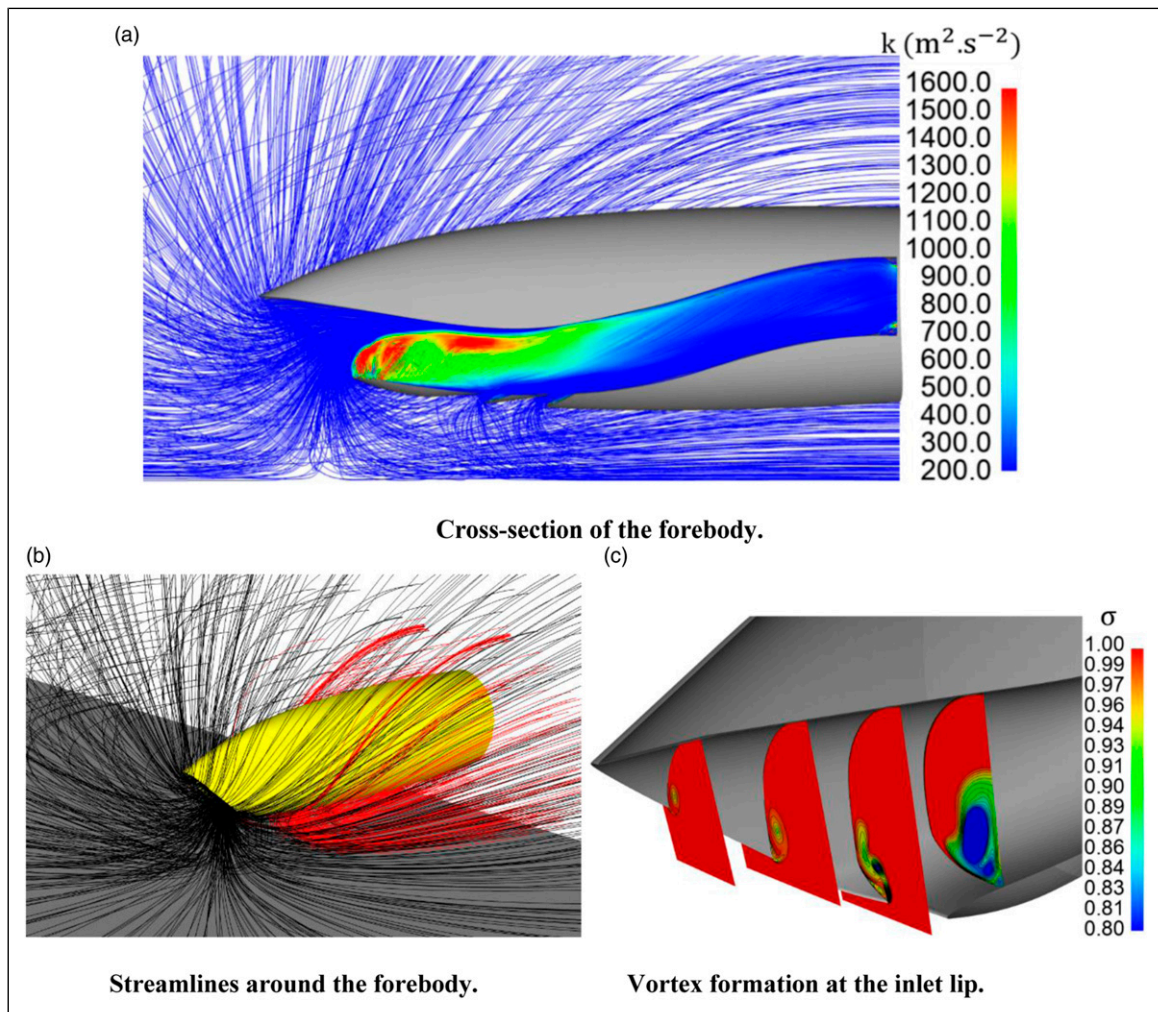


Figure 20. Inlet flow structure at static operation. (a) Cross-section of the forebody. (b) Streamlines around the forebody. (c) Vortex formation at the inlet lip.

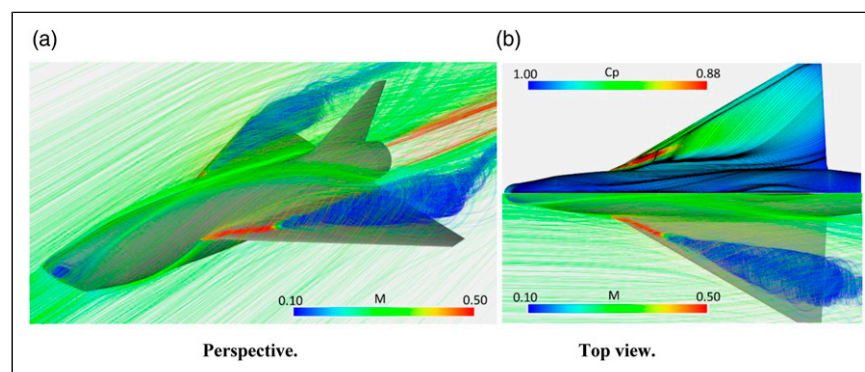


Figure 21. Drone flow structure at $M_\infty = 0.30$ and $AoA = 20^\circ$ (transparent streamlines). (a) Perspective. (b) Top view.

concept from Ref. (37) is designed for a supersonic executive jet. It uses an isentropic compression surface with a short subsonic diffuser duct similar to the S3DT inlet.³⁹ Results show that the viscous loss on the long S-duct diffuser imposes more than 2% pressure recovery loss to the AIP. According to Ref. (35), the basic distortion index

(DC_θ) is calculated to indicate the level of compatibility between the inlet and the compressor.

In order to illustrate the impact of the curved duct of the drone on the flow quality, the flow structure of the inlet with a straight duct is also simulated for comparison. The detailed discussions about the measured distortion level

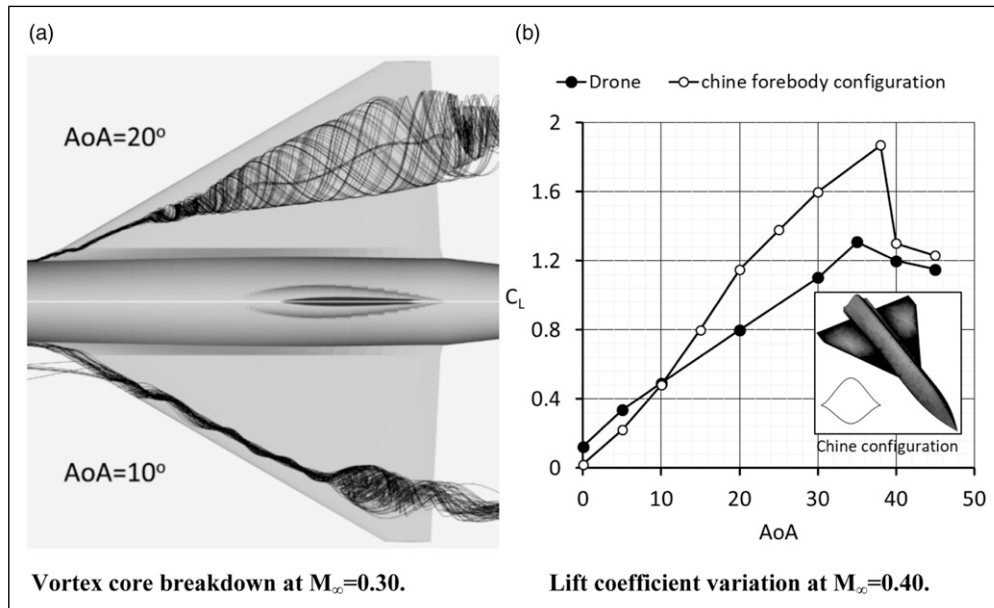


Figure 22. Wing flow structure and lift behavior. (a) Vortex core breakdown at $M_\infty=0.30$. (b) Lift coefficient variation at $M_\infty=0.40$.

can be found in Ref. (34). The index values at $M = 1.6$ are compared with a supersonic inlet of a conceptual fighter jet⁴⁰ in Figure 18b, and Figure 18c shows the result at subsonic speeds with the same behavior. Although the resulting pressure recovery is still close to MIL-STD in Ref. (10), reducing boundary layer thickness at the downstream end of the duct is necessary. Figures 19a,b show the impact of boundary layer bypass on the AIP flow pattern. Pressure recovery has been improved to 0.967 at the design speed which matches the MIL-STD. Table 11 summarizes the efficiency factors for the new duct.

At static condition and without the auxiliary doors, the captured mass flow rate at compressor face is about 71% of maximum engine demand. The static operation of the inlet is demonstrated by the captured streamlines in Figure 20a when the auxiliary doors are open; the streamlines are colored by turbulence kinetic energy (k) which illuminate a highly turbulence pattern at the entrance. The red streamlines in Figure 20b are captured by the auxiliary windows which increases the captured mass flow up to 83%. The vortex formation is demonstrated by pressure ratio contours in Figure 20c. This chaotic pattern loses its vorticity along the duct and the k level reduces to below $200 \text{ m}^2 \cdot \text{s}^{-2}$ before the compressor face when Mach number at the compressor face is about 0.5. The auxiliary doors do not have a significant impact on the vortical flow pattern but increase the captured mass flow up to 8%.

External aerodynamics

At subsonic speeds, delta wings produce a vortex pair on the upper surface at moderate to high angles of attack.^{41–44} Delaying the stall at high AoA is the main advantage of these vortices⁴⁴ which are shown in Figures 21a,b at $M = 0.3$ and $AoA = 20^\circ$. The streamlines also indicate a pair of

counter rotating vortices over the forebody which are generated from the separation bubble at the nose. At level flight condition, no vortices are detected on the fuselage. The weak vortices from the nose pass along the top of the fuselage and diverge before reaching the vertical stabilizer. Such a safe passage has been observed at all tested (low subsonic) Mach numbers. The oil pattern in Figure 21b illustrates the trace of the nose vortices on the fuselage clearly. Figure 22a shows the effect of AoA on the vortex core breakdown along the leading edge. Generally, at $AoA > 20^\circ$, the breakdown starts near the wing root and a large vortical expansion covers the upper surface of the wing.

The resulting stall characteristic of the concept can be demonstrated by comparing the lift coefficient variation in Figure 22b. In this figure, the wing of chine configuration without the inlet⁴⁵ has a symmetric airfoil with sharp leading and trailing edges. After $AoA = 35^\circ$, a smooth change in curve is measured for the drone, but this change for the chine configuration is in the form of a sudden jump.

Another pair of vortices is detected under the wing at high AoA , especially at high-subsonic regimes. Figure 23a illustrates their trace on the fuselage by comparing the oil patterns. These weak vortices are captured by tangential streamlines on a cross-sectional Mach contour in Figure 23b. The phenomenon can be explained by considering the effects of the high pressure zone under the wing in rolling up the incoming boundary layer at high AoA . Figure 23c shows the Mach contour on the midwing cross-section at $AoA=0^\circ$, and Figure 23d illustrates the pressure distribution along the longitudinal cross-sections on the wing. As it can be seen, the local Mach number over the wing is below 1.0 at $M_\infty=0.8$; therefore, the selected airfoil could delay the normal shock formation at cruise back to base successfully.

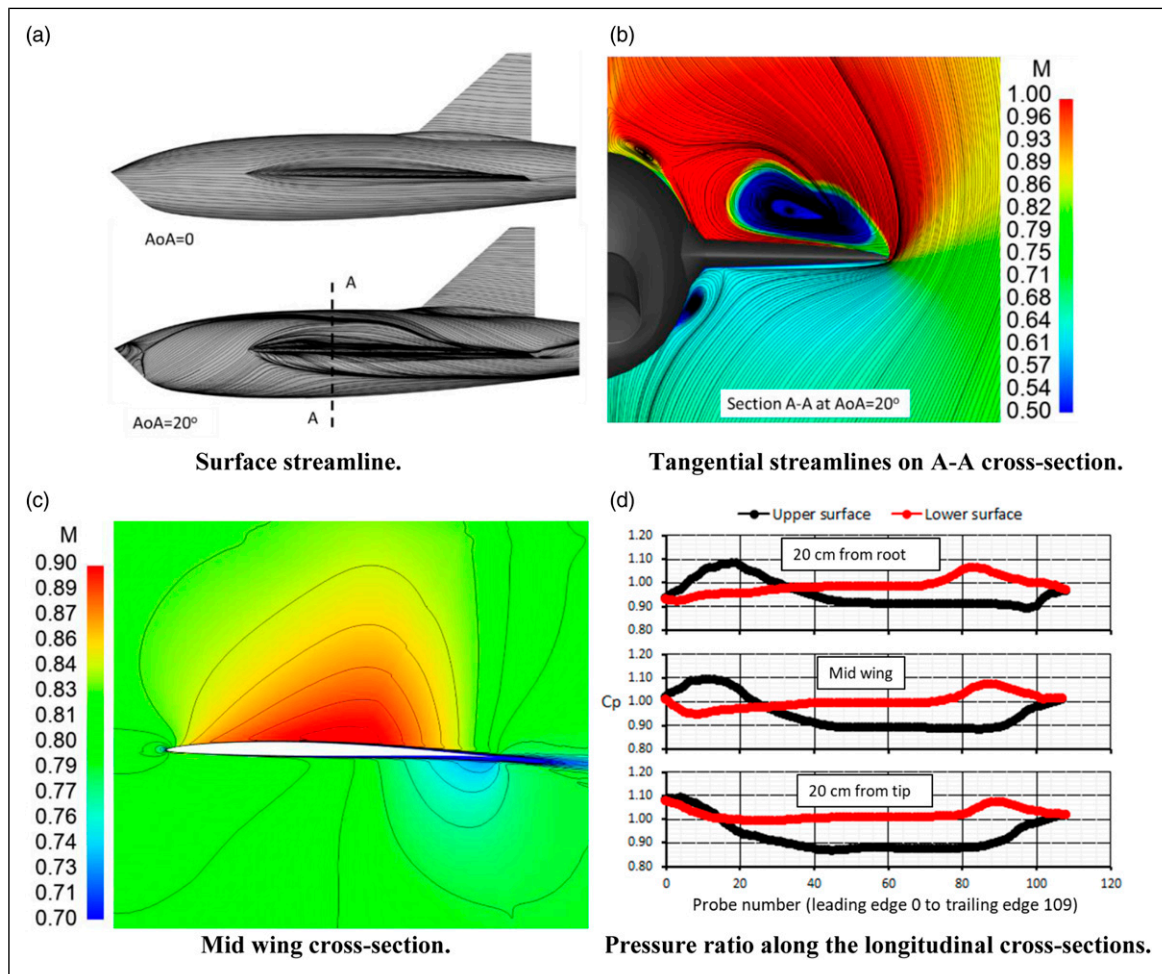


Figure 23. Flow structure at cruise back flight, $M_\infty=0.8$. (a) Surface streamline. (b) Tangential streamlines on A-A cross-section. (c) Midwing cross-section. (d) Pressure ratio along the longitudinal cross-sections.

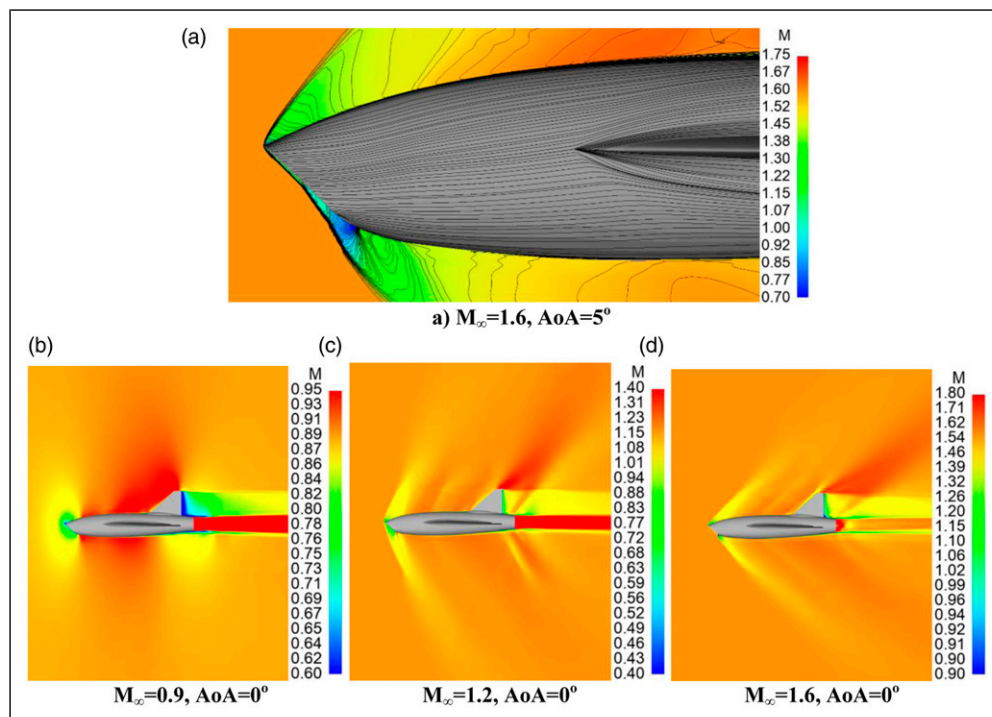


Figure 24. Flow pattern on the plane of symmetry. (a) $M_\infty=1.6, AoA=5^\circ$. (b) $M_\infty=0.9, AoA=0^\circ$. (c) $M_\infty=1.2, AoA=0^\circ$. (d) $M_\infty=1.6, AoA=0^\circ$.

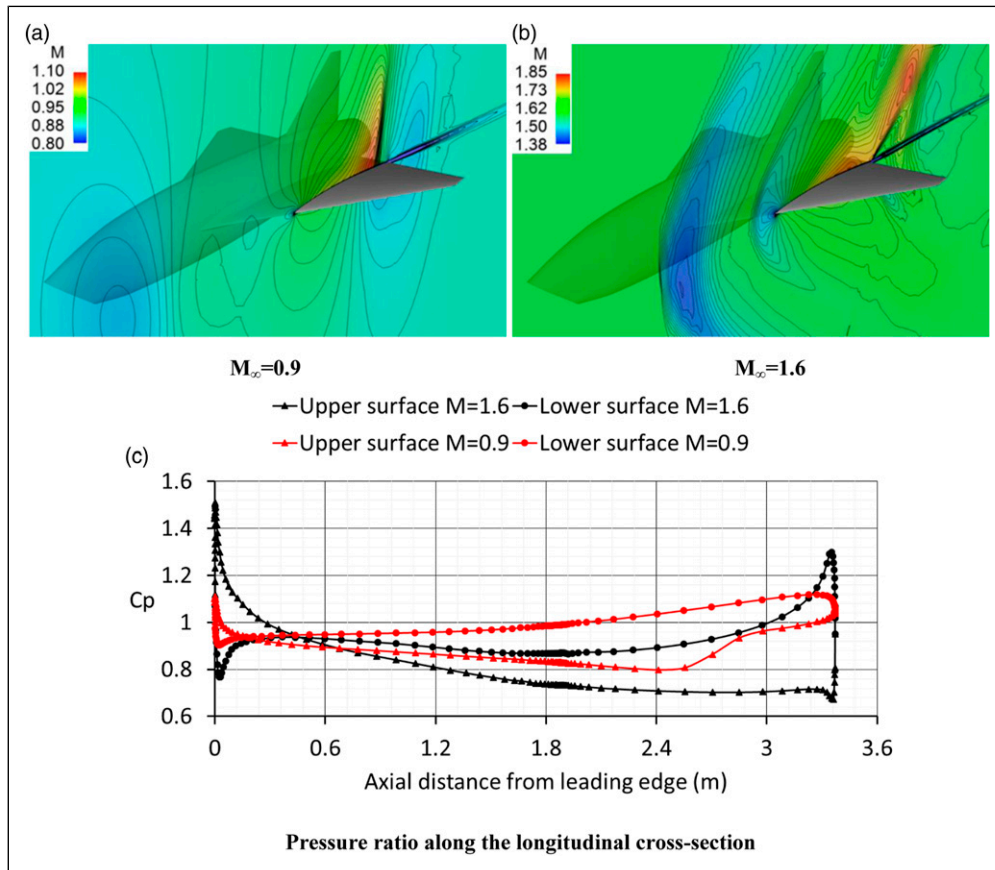


Figure 25. Wing flow structure on a cross-section. (a) $M_\infty=0.9$. (b) $M_\infty=1.6$. (c) Pressure ratio along the longitudinal cross-section.

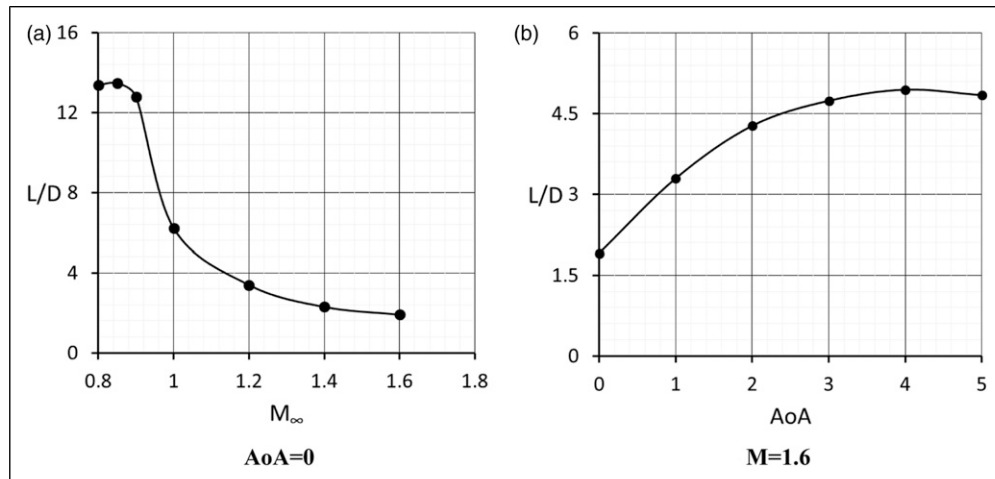


Figure 26. Drone L/D variation. (a) $AoA=0$. (b) $M=1.6$.

For the Mach range $0.9 < M < 1.2$, the local shock waves on the fuselage may disturb the boundary layer, but at $M=1.6$ and $AoA < 5$, no flow separation on the forebody is observed. This consistent flow pattern on the forebody can be seen in Figure 24a.

Figures 24b–d show the transonic to supersonic flow patterns around the drone by Mach contours on the plane of symmetry. At $M_\infty \leq 0.8$ and $AoA=0^\circ$, no local shock waves appear on the external surfaces, the flow structure is uniform around the forebody, and other aerodynamic

surfaces receive no vortices from the nose section. The wing's flow pattern at $M_\infty=0.9$ and $M_\infty=1.6$ is shown by Mach contours in Figures 25a,b on a streamwise vertical section.

At $M_\infty=0.9$, a weak normal shock appears near the trailing edge which is recognizable in Figure 25a. Figure 25c shows this shockwave by the pressure rise on the upper surface pressure distribution.

The effect of Mach number and AoA on L/D is plotted in Figures 26a,b; it can be seen that the maximum

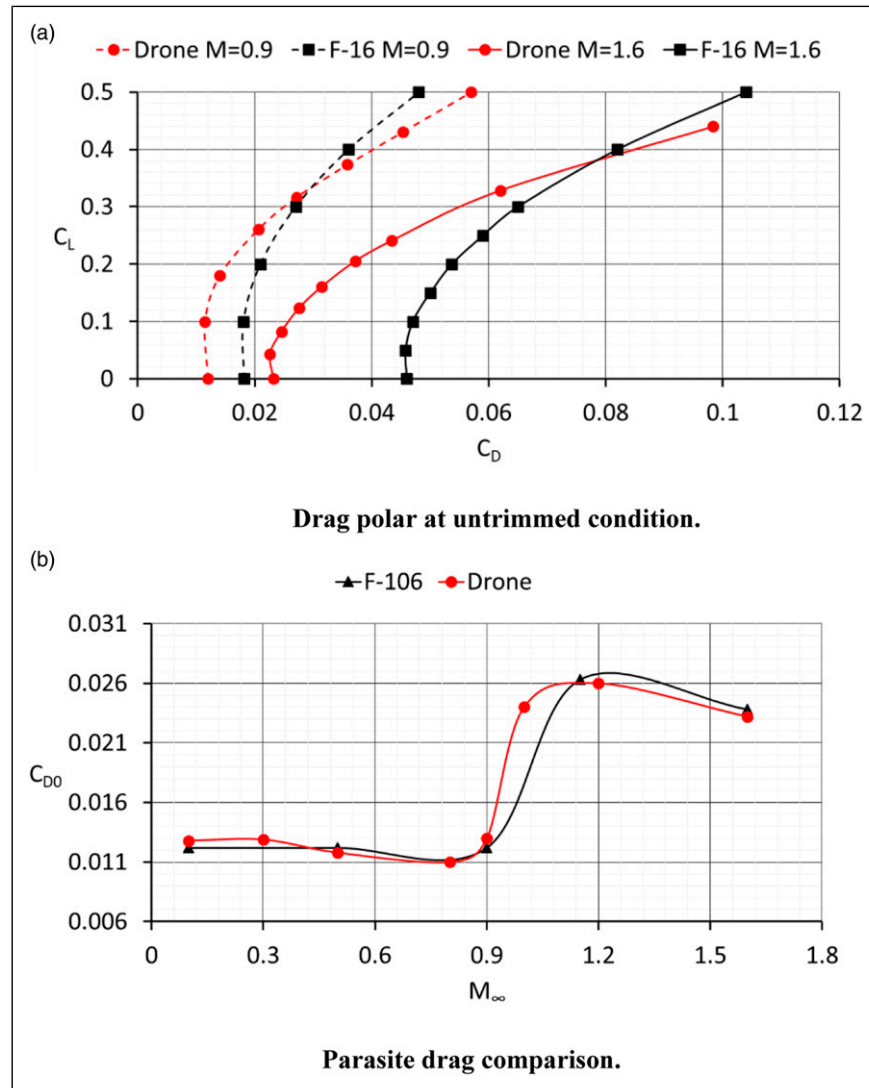


Figure 27. Drag comparison diagrams. (a) Drag polar at untrimmed condition. (b) Parasite drag comparison.

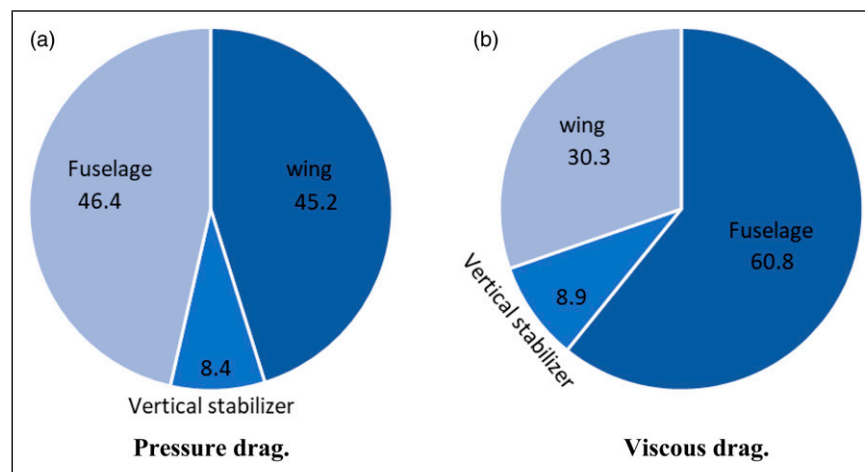


Figure 28. Drag breakdown at $M_\infty = 1.6$ (percentage). (a) Pressure drag. (b) Viscous drag.

supersonic L/D at $M_\infty = 1.6$ is generated at $AoA = 4^\circ$ which does not exceed the inlet pitch-up limit. The transonic to supersonic aerodynamic efficiency of the concept is compared to F-16⁴⁶ by the drag polar diagrams in Figure

27a, and the parasite drag variation is compared by F-106 (from Ref. 17) in Figure 27b. The impact of the area-ruled fuselage of the F-106 can be seen with lower C_{D0} at $0.9 < M < 1.2$ in Figure 27b. The resulting drag of the

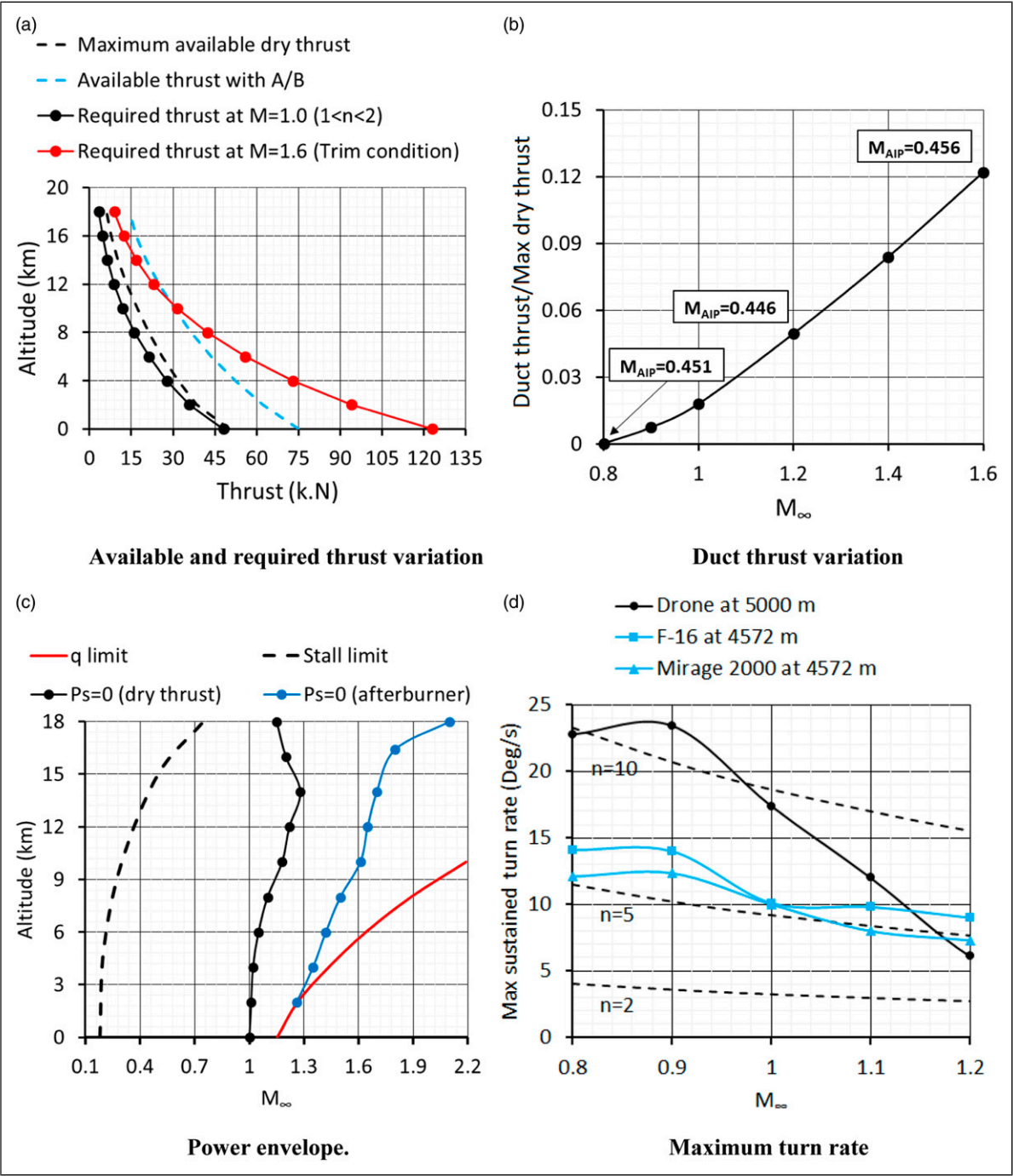


Figure 29. Performance estimation diagrams. (a) Available and required thrust variation. (b) Duct thrust variation. (c) Power envelope. (d) Maximum turn rate.

Table 12. Comparing sea level thrust and max speed.

Airplane	Maximum sea level thrust (k.N)	M_{max} at sea level	L/D_{max}	Transonic C_{D0}
Drone	75	1.1	12.8	($M_\infty=0.8$) 0.0109
Republic F-105D ⁴⁹	117	1.1	10.4	0.0173
Vought F-8H ⁴⁹	80	1.0	12.8	0.0133
Convair F-106A ⁴⁹	109	—	12.1	0.0083

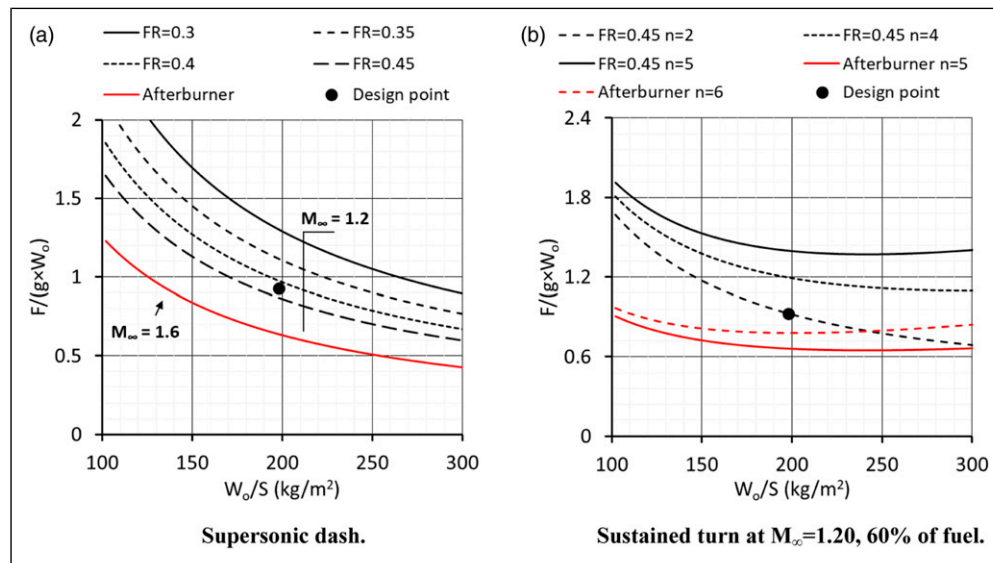


Figure 30. Final design point on initial constraint diagrams (10 km). (a) Supersonic dash. (b) Sustained turn at $M_\infty=1.20$, 60% of fuel.

drone's aerodynamic surfaces at design speed are compared in Figures 28a,b. More than 46.4% of supersonic pressure drag at $AoA=0^\circ$ is generated by the fuselage while the viscous drag forms about 22% of total drag.

Performance

To calculate the performance of the drone, a thrust model is developed by GasTurb⁴⁷ software based on the inlet mass flow map at different flight altitudes (sea level to 18 km) and a typical twin-spool low bypass turbofan engine with pressure ratio of 24.5, similar to the Snecma M-88. Figure 29a compares the resulting thrust variation of the simulated engine and the required thrust for speed limits based on CFD simulations. At Mach 1.6, the pressure distribution on the duct surface generates a considerable thrust force which overcomes 20% of the overall drag. The variation of duct thrust with Mach number is plotted in Figure 29b. Figures 29c,d represent the results of the energy-maneuvering study for the concept based on specific excess power (Ps) variation and maximum sustained turn rates. Figure 29c indicates that the drone can easily accelerate through $M_\infty=1.0$ and reach $M_\infty=1.2$ without afterburner at altitudes above 10 km. Even including trim drag as 5% of the total drag (when $M > 1.0$), the system can accelerate above Mach 1.6 with an afterburner which is matched with the mission requirement. Comparing the maximum level turn rate of the concept (at $Ps=0$) with two high-production fighters⁴⁸ in Figure 29d demonstrates the superiority of the drone's transonic aerodynamics. At the sea level, accelerating through Mach 1.1 is possible with the afterburner and a clean configuration. Table 12 compares the drone's sea level speed with other combat aircraft.

The impact of the afterburner on the design solution can be seen on the constraint diagram in Figure 30a, which is plotted for different thrust ratios ($FR = \frac{\text{Available thrust}}{\text{Maximum thrust}}$). The standard A/B thrust of the Snecma M-88 is more than dash

requirements (by assuming 35–40% thrust gain), but as Figure 30b shows, the drone is incapable of supersonic sustained turns at $n > 2$ at an altitude of 10 km without thrust augmentation.

Conclusion

This work demonstrates the aerodynamic efficiency of a baseline supersonic drone concept that can meet the defined mission requirements based on an available 50 kN thrust level engine. After determining the mission requirements, the conceptual design phase is started by weight estimation of the system and then an open-nose configuration is selected to meet the mission requirements, especially the sustained supersonic flight. Finally, the aerodynamic layout of the concept is generated based on internal volume constraints and its aerodynamics is studied using numerical simulations which indicate that forebody-inlet configuration is highly efficient and provides the required mass flow and low distortion level at all flight phases. Although the subsonic diffuser is highly curved, the flow quality indices are in the required range and pressure recovery over 0.97 at $AoA=0^\circ$ is reachable by using a concentric boundary layer bleed gap at the end of the diffuser. The baseline pressure recovery (without boundary layer gap) varies between 0.98 at low subsonic speeds and 0.95 at $M_\infty=1.6$. At static operation, when the average Mach number at the compressor face is 0.45–0.50, the resulting mass flow rate will be about 71% of maximum engine demand. Increasing the captured mass flow is investigated by adding the auxiliary doors which improves the captured mass up to 83% of maximum demand. The flow separation at the cowl lip is the dominant phenomenon at static operation which results in a chaotic flow pattern at the entrance and the blow-in doors cannot improve the flow distortion at the entrance but the vorticity magnitude drops sharply along the duct. One important

difference between the classic open-nose forebodies and the proposed concept is the location of the radar which is behind the compression surface, so the usage of composite material is unavoidable for the nose.

The forebody generates no vortices at level flight, so that the wing root receives a uniform flow pattern. At moderate to high AoAs, the flow separation region appears on the nose (over the sharp lip of the compression surface) which forms a pair of vortices. These weak vortices deform smoothly along their paths between the wing and vertical stabilizer, and no breakdown is detected. This configuration needs to fly with high AoA when $M_\infty < 0.25$ because of the low gradient of the C_L vs. α curve. Although the baseline fuselage is not area-ruled, the drag polar comparison shows that the open-nose concept is highly efficient at the supersonic design point, and its zero-lift drag coefficient is about 0.0226, which is lower than that of the F-106. Exceeding $M=1$ without afterburner is possible at altitudes greater than 10 km but reaching the design point requires the afterburner. Maximum sustained turn rate at 5 km is about 23 deg/s which is higher than comparable conventional configurations. Because there is no pilot, the flight vehicle can perform sustained high-g maneuvers at transonic speeds. Because of rather simple and robust geometry, there is some space for additional optimization; changing the blow-in doors position, adding a canard to improve the maneuverability at low subsonic speeds, and replacing the current classic delta wing with a more advance design are topics of future aerodynamic studies.

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ORCID iDs

Eiman B. Saheby  <https://orcid.org/0000-0001-7153-3427>

Shen Xing  <https://orcid.org/0000-0001-5002-3030>

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Appendix

α = angle of attack

AIP = aerodynamic interface plane

A/B = afterburner (reheat)

BPC = base-plane curve

E, F, G = inviscid flux vectors

E_v, F_v, G_v = viscous flux terms

FR = available thrust to maximum sea level thrust

k = turbulence kinetic energy

k_{ave} = averaged turbulence kinetic energy at AIP

M = Mach number

M_∞ = freestream Mach number

M_{out} = Averaged Mach number at the outlet

$\dot{m}_{AIP}$ = calculated mass flow rate at the AIP

MFR = mass flow ratio

M_{bypass} = averaged Mach number at bypass

MIL-STD = United States Department of Defense standard

N-S = Navier-Stokes

SST = shear stress transport

P = static pressure

P_t = total pressure

P_{t_∞} = freestream total pressure

P_{tAIP} = mean total pressure at the AIP

P_{out} = static pressure of outlet boundary condition

q = variable vector

S = source term

α = angle of attack

β = sideslip angle

ω = longitudinal vorticity

σ = total pressure ratio